

JANIT B

Salem, Tamil Nadu, India | +91 63802 86204 | janit.b2006@gmail.com | linkedin.com/in/janit2006 | github.com/JNT-h04

SUMMARY

Final-year B.Tech CSE (AI & ML) student who takes machine-learning systems the whole way: data, training, honest evaluation, API and interface. Four models shipped as working web demos behind one tested FastAPI service, each reported with its per-class accuracy, failure modes and leakage checks — not just a headline number.

EDUCATION

Vellore Institute of Technology, Chennai — B.Tech CSE (AI & ML) 2023 – 2027
CGPA 8.24 / 10

Sri Chaitanya School, Salem Till 2023
Class 12: 83.8% | Class 10: 86.2%

EXPERIENCE

AI & Software Engineering Intern — Strategy Fox International Pvt. Ltd. May – June 2026
Remote / Bangalore — AI-automation studio for D2C and retail brands

- Built a real-time AI voice agent with ElevenLabs that handled live customer conversations for client brands, removing manual follow-up the team had done by hand.
- Wired conversational flows to REST APIs with dynamic variables, so the agent could look up live order and customer records mid-call rather than read a fixed script.
- Shipped features and fixes on client Shopify storefronts, working with engineering and business stakeholders from requirement to deployed feature.

PROJECTS

Explainable Alzheimer's MRI Staging PyTorch, ResNet18/50, Grad-CAM, FastAPI

- Found a data-leakage flaw in the original approach: random slice splits scored ~99% but put the same patients in train and test. Re-splitting by patient gave the true 4-class baseline, **28.7%**.
- Shipped a validation-weighted ResNet18 + ResNet50 ensemble: **60.2%** accuracy (macro F1 0.59) on 693 held-out slices, and **81.5%** per patient across 54 held-out patients by voting a patient's slices.
- Excluded a class holding only two patients rather than report an unsupportable number, and exposed Grad-CAM heatmaps so every prediction can be inspected.

AI Meeting Assistant Gemini, FastAPI, background jobs, React

- Designed an upload-and-poll pipeline (HTTP 202 with a job id, background worker, status polling) so transcribing a long recording outlives a browser request timeout.
- Constrained the model to a typed JSON schema to return structured minutes — summary, decisions, owner-tagged actions and a speaker-labelled transcript — with rate limiting and deletion of the upload once the job ends.

Structural Crack Severity Detection TensorFlow, ResNet50, OpenCV, FastAPI

- Fine-tuned ResNet50 on **30,015** concrete images: **96.9%** validation accuracy, 92 of 100 correct on a random held-out sample.
- Paired the classifier with classical computer vision (Canny edge density, Hough line angles) to report crack intensity, likely cause and maintenance guidance, documenting the weakest class rather than quoting one figure.

Book Summariser with Streaming Output Gemini, PyMuPDF, FastAPI, React

- Parses PDF, DOCX, EPUB and text, detecting chapters from document headings rather than a fixed page count, and streams each summary, key insight and flashcard token by token with model fallback.

Cloud Resource Optimisation with Reinforcement Learning PyTorch, LSTM, PPO, AWS

- Paired LSTM workload forecasting with a PPO agent that learns scaling actions from reward instead of fixed thresholds, benchmarked against reactive and predictive strategies on cost versus SLA compliance.

Portfolio Platform hosting the demos above React, TypeScript, FastAPI, Pytest

- One FastAPI service serving all four models with background model loading, per-visitor rate limiting and **46 passing tests**, behind two interchangeable front-end themes that share one logic layer.

TECHNICAL SKILLS

Languages Python, TypeScript, Java, C++, SQL

ML & Deep Learning PyTorch, TensorFlow / Keras, scikit-learn, CNN / ResNet, LSTM, Transformers, Hugging Face, Grad-CAM

LLM & RAG Gemini API, LangChain, FAISS, Whisper, prompt engineering, structured JSON output

Backend & Web FastAPI, REST APIs, streaming responses, background jobs, Pytest, React, Vite, Streamlit

Tools & Platforms Git & GitHub, AWS, Docker (basics), OpenCV, Linux, ElevenLabs, Shopify

Foundations Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

IBM Generative AI with watsonx — VIT (2025)